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**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks: 50

Annexure A

Analytical Problem Solving

Code of Conduct:

I V. Murali Karabik (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V. Murali Karabik
Signature of the student:

level - 3

FP

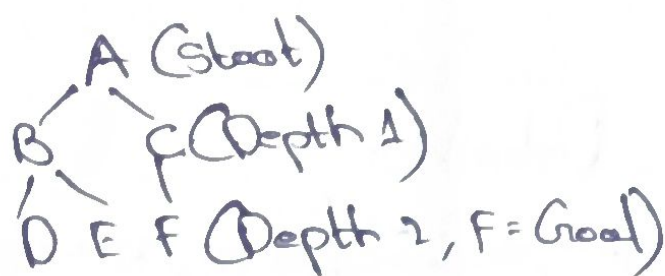
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Explain Iterative Deepening Search (IDS) and trace its execution on a graph of your choice with at least 6 nodes, showing the search at depth limits 0, 1, 2. Discuss why IDS combines the space η of DFS with the completeness of BFS.

Explanation:-

IDS combines space η of DFS with completeness of BFS. It performs depth limited searches with incrementally increasing depth limits (0, 1, 2...) until goal found.

Graph Visualization:-



Adjacency List

A: [B, C] C: [F]

B: [D, E] D, E, F: []

IDS Execution trace (Finding F):-

Depth	Nodes Explored	Result.
0	A	Not Found
1	A, B, C	Not Found
2	A, B, D, E, C, F	SUCCESS of F

Detailed trace Analysis:-

Depth 0:- Only root A checked, Goal F not at depth 0

Depth 1:- A expanded to B, C. Neither is F.

Depth 2:- A $\rightarrow$ B $\rightarrow$ D (Leaf), A $\rightarrow$ C $\rightarrow$ F (Goal Found!).

Key Insight: IDS explores all nodes at depth d before $d+1$ (like BFS), avoiding large memory requirements. Redundant node exploration is minimal cost for huge memory savings.

- 2) A vacuum-cleaner robot operates in a two-room environment (Room A & Room B), each of which can be clean or dirty. Define the PEAS (Performance measure, Environment, Actuators, Sensors) description for this agent and characterize its task environment along the dimensions of observability, determinism, episodicity, and no. of agents.

A) PEAS Description:

Component	Details.
Performance.	+1/clean square, -1/action, maximize η
Environment	Two rooms (A, B), each clean or dirty
Actuators.	Move left, Move Right, Suck, Idle
Sensors	Location sensors, Dirt sensor, Battery sensor.

Task Environment Characterization:-

Dimension	Type	Justification.
Observability	Fully observable	Complete state info: location, battery
Determinism	Deterministic	Actions have predictable outcomes
Episodicity	Sequential	Decisions depend on history
Agents	Single-Agent	Only one robot, no multi-agent

depth. An e-commerce company wants to deploy a delivery drone to autonomously deliver packages within a city. Identify what type of problem solving agent is suitable for this scenario, and write pseudocode outlining how the agent formulates and solves the delivery routing problem.

A) Suitable agent :- Goal-based problem solving agent using Search.

PseudoCode :-

- Perceive package and destination.
- Formulate goal = deliver package.
- Build city map graph.
- Run A* to compute shortest safe route.
- Follow route.
- Replan if obstacle/weather changes.
- Deliver package and confirm completion.

Example graph used for IDS :-

